

Algebra 1
Unit 1
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

1. Which of the following is not a radical expression that is equivalent to $\sqrt{108}$?

- A. $\sqrt{2} \cdot \sqrt{54}$
- B. $\sqrt{3} \cdot \sqrt{36}$
- ☒ C. $\sqrt{5} \cdot \sqrt{21}$
- D. $\sqrt{6} \cdot \sqrt{18}$

2. Which of the expressions from #1 can be rewritten in the form of an integer multiplied by the radical? *Explain.*

B. $\sqrt{3} \cdot \sqrt{36}$

Because $\sqrt{36}$ is 6 which is an integer.

For questions 3-4, write two equivalent expressions for each radical expression using the product law of radicals. One of your expressions should simplify to an integer multiplied by a radical.

3. $\sqrt[3]{200} = \sqrt[3]{2} \cdot \sqrt[3]{100}$

$\sqrt[3]{200} = \sqrt[3]{8} \cdot \sqrt[3]{25} = 2\sqrt[3]{25}$

4. $\sqrt{72} = \sqrt{9} \cdot \sqrt{8} = 3\sqrt{8}$

$\sqrt{72} = \sqrt{36} \cdot \sqrt{2}$

5. Rewrite $\sqrt[3]{200}$ as an integer multiplied by a radical.

$2\sqrt[3]{25}$

6. Rewrite $\sqrt{72}$ as an integer multiplied by a radical.

$6\sqrt{2}$

7. Identify all expressions that are equivalent to $-4\sqrt[3]{48}$.

- $2\sqrt[3]{6}$
- $-8\sqrt[3]{6}$ ✓
- $-8\sqrt[3]{12}$
- $32\sqrt[3]{2}$
- $-4 \cdot \sqrt[3]{8} \cdot \sqrt[3]{6}$ ✓
- $-4 \cdot \sqrt[3]{12} \cdot \sqrt[3]{4}$ ✓
- $-4 \cdot (24 \cdot 2)^{\frac{1}{3}}$ ✓

For questions 8-10, write an equivalent expression in the form of a constant multiplied by a radical.

8. $\sqrt{392} = \sqrt{196 \cdot 2} = \boxed{14\sqrt{2}}$

9. $\sqrt[3]{135} = \sqrt[3]{27 \cdot 5} = \boxed{3\sqrt[3]{5}}$

10. $-3\sqrt[3]{625} = -3\sqrt[3]{125 \cdot 5} = -3(5\sqrt[3]{5})$
 $= \boxed{-15\sqrt[3]{5}}$